

API Endpoint Table

The following table outlines the RESTful API routes implemented in the backend.

Category	Method	Endpoint	Access	Description
Auth	POST	/api/auth/register	Public	Register a new student account.
Auth	POST	/api/auth/login	Public	Authenticate user and return JWT.
Auth	GET	/api/auth/users	Admin	List all registered users.
Auth	PATCH	/api/auth/profile	All	Update own phone or password.
Rooms	GET	/api/rooms	All	List all rooms and their status.
Rooms	POST	/api/rooms	Admin	Create a new room with image.
Rooms	PATCH	/api/rooms/:id	Admin	Update room details/status.
Rooms	DELETE	/api/rooms/:id	Admin	Delete a room from the system.
Bookings	POST	/api/bookings	Student	Request a room allocation.
Bookings	GET	/api/bookings	All	View own/all booking requests.
Bookings	PATCH	/api/bookings/:id/approve	Admin	Approve a booking and update occupancy.
Maint.	POST	/api/maintenance	Student	Report a facility issue.
Maint.	GET	/api/maintenance	All	Track status of requests.
Maint.	PATCH	/api/maintenance/:id	Admin	Update status (e.g., In-Progress).
Visitors	POST	/api/visitors	Admin	Register a new visitor.
Visitors	GET	/api/visitors	All	List all visitor logs.
Fees	POST	/api/fees/payment	All	Upload fee payment proof.
Fees	GET	/api/fees/payments	All	View payment history.
Fees	PATCH	/api/fees/payments/:id/approve	Admin	Mark a payment as 'Paid'.

Technical Implementation Notes:

- **Response Format:** All endpoints return standard JSON.
- **Authentication:** Protected routes require an Authorization: Bearer <token> header.
- **Error Handling:** Consistent use of status codes (200, 201, 400, 401, 403, 404, 500).